

Question

M is a silver-gray, hard and brittle transition metal with metallic luster. It is the main additive of stainless steel, which can make steel have high corrosion resistance. The reaction of nitric acid and **M** amalgam can obtain a deliquescent black oxide powder **A**, which can burn in the air to obtain **B**. A 1:1 mixture of carbon monoxide and chlorine gas at high temperature can convert **B** into **C**. The fusion reaction of **B** with potassium chlorate can obtain a yellow solid **D**. After dissolving **D** in water and acidifying, the solution will turn orange due to the formation of **E**. Further addition of concentrated sulfuric acid to the solution will form an orange-red solid **F**.

It is known that there is the following process:

- C** can react with Ph_4AsCl and sodium azide under certain conditions to generate a mononuclear complex, in which the mass fraction of nitrogen is 17.34%.
- Powdered **E** is heated to boiling with excess concentrated hydrofluoric acid and then cooled to obtain a substance **G** containing **M** 32.89%.
- E** reacts with **C** under molten conditions, and one of the products is similar to **G**.

The following statement is correct:

- A.** All other options are incorrect
- B.** The mass fraction of **M** in **A** is 68.4%
- C.** The mass fraction of **M** in **C** is 33.6%
- D.** The mass fraction of **M** in **F** is 76.5%
- E.** The complex in Process 1 is a 1:1 type ionic compound.

- F.** In Process 1, the hybridization of **M** in the complex is sp^3 hybridization.
- G.** The sum of the coefficients of all species in the chemical equation for Process 3, balanced to the simplest integer ratio, is 12.
- H.** The sum of the coefficients of all species in the chemical equation of Process 2, when balanced to the simplest integer ratio, is 8.

Answer

Correct Answer: **G**

Detailed Explanation

M is the main additive of stainless steel, and is hard and brittle, generally Ni or Cr.

CHECKPOINT

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M is the main additive of stainless steel, generally Ni or Cr

The yellow solid **D** dissolves in acidic solution to produce an orange color. Based on element knowledge, it is easy to think of chromate CrO_4^{2-} and dichromate $\text{Cr}_2\text{O}_7^{2-}$. Therefore, **M** is Cr, **D** is potassium chromate K_2CrO_4 , and **E** is potassium dichromate $\text{K}_2\text{Cr}_2\text{O}_7$.

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M is Cr

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D is K_2CrO_4 , **E** is $\text{K}_2\text{Cr}_2\text{O}_7$

A can burn to produce **B**, and **B** can be fused with potassium chlorate to produce K_2CrO_4 , then **B** should be the stable oxide Cr_2O_3 . At this time, **A** can only be the low-valent oxide CrO .

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B should be Cr_2O_3

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A is CrO

Cr_2O_3 is converted to **C** by 1:1 carbon monoxide and chlorine gas, so **C** should be the chloride CrCl_3 , and CrO_2Cl_2 will not be generated under reducing conditions.

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C is the chloride CrCl_3

The orange-red solid **F** produced by adding concentrated sulfuric acid to the $\text{K}_2\text{Cr}_2\text{O}_7$ solution is CrO_3 .

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F is CrO_3

According to the chemical formula, options B, C, and D are all incorrect.

The following studies three processes:

Process 1, Ph_4AsCl and sodium azide form a complex with CrCl_3 . According to the Lewis acid-base reaction, the cation is likely to be Ph_4As^+ . The species that can be used for coordination in the system are likely to be N_3^- or Cl^- and can only be monodentate ligands. Considering that Cr(III) is commonly hexacoordinate, the chemical formula of this mononuclear complex may be $(\text{Ph}_4\text{As})_3[\text{Cr}(\text{N}_3)_x\text{Cl}_{6-x}]$. Considering the mass fraction of nitrogen, try the specific value of x . When x is 6, it satisfies the mass fraction of nitrogen 17.34% given in the question. Therefore, the chemical formula of the complex is $(\text{Ph}_4\text{As})_3[\text{Cr}(\text{N}_3)_6]$, which is a 3:1 type ionic compound, and option E is incorrect. Cr(III) is d^2sp^3 hybridized, and option F is incorrect.

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The cation is likely to be Ph_4As^+

CHECKPOINT

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Cr(III) is commonly hexacoordinate, the chemical formula of this mononuclear complex may be $(\text{Ph}_4\text{As})_3[\text{Cr}(\text{N}_3)_x\text{Cl}_{6-x}]$

CHECKPOINT

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The chemical formula of the complex is $(\text{Ph}_4\text{As})_3[\text{Cr}(\text{N}_3)_6]$

CHECKPOINT

1 PTS

Cr(III) is d^2sp^3 hybridized

Process 2, $K_2Cr_2O_7$ reacts with HF to obtain a substance **G** containing Cr 32.89%. Since HF does not exhibit reducing properties, this reaction is not a redox reaction, so **G** is likely to be a fluoro complex. Consider the chemical formula of the complex as CrO_xF_y , and containing Cr(VI). Common compounds may be $KCrO_3F$, CrO_2F_2 and $KCrOF_5$. After calculation, it can be seen that $KCrO_3F$ satisfies the mass fraction given in the question. Therefore, **G** is $KCrO_3F$, balance the chemical equation:

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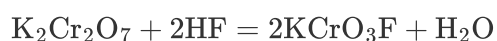
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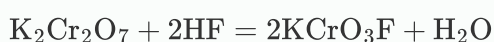
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G is $KCrO_3F$



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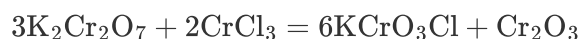
The sum of the coefficients is 6, and option H is incorrect.

Process 3: $\text{K}_2\text{Cr}_2\text{O}_7$ reacts with CrCl_3 in a molten state. The product is similar to KCrO_3F , so it is obvious that the product is KCrO_3Cl . Under molten conditions, a stable oxide Cr_2O_3 should be generated, so balance the equation:

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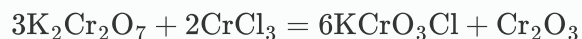
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The products are KCrO_3Cl and Cr_2O_3



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The sum of the reaction coefficients is 12, and option G is correct.